

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

HFq3RNHTGrB6yBgtwwZuLjEmGSr942PcSNUXVF46ULTD

Date: 5/1/2025

Compounds: Amoxicillin, Aspirin, Bismuth Subsalicylate, Cetirizine

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: AmoCetiBiSPirin

Formula: $C_{21}H_{27}N_3O_7 + C_9H_8O_4 + Bi(C_7H_5O_4)_3 + C_{21}H_{25}ClN_2O_3$

Weight: 1071.51

Drug-likeness: 0.67

Activity:

Antibiotic, Anti-inflammatory, Gastroprotective, Antihistamine

Target Analysis

Bacterial cell wall synthesis enzymes

Binding Affinity: -9.5

Critical for bacterial infection treatment

Cyclooxygenase (COX 1 and 2)

Binding Affinity: -7.8

Anti-inflammatory and analgesic effects

Histamine H1 receptor

Binding Affinity: -8.2

Relief from allergy symptoms

Toxicity Profile

Risk Level: Moderate

Affected Systems: Liver, Kidneys, Gastrointestinal tract

Mitigation Suggestions:

- Regular liver function tests
- Adequate hydration
- Use gastroprotection

Pharmacokinetics

Absorption:

Rapid oral absorption

Distribution:

Widely distributed, crosses blood-brain barrier

Metabolism:

Hepatic metabolism via cytochrome P450

Excretion:

Renal and fecal excretion

Half-life: 4-6 hours

Detailed Analysis

AmoCetiBiSPirin is an advanced hypothetical drug compound designed to integrate the therapeutic advantages of Amoxicillin, Aspirin, Bismuth Subsalicylate, and Cetirizine. This compound aims to provide a combined antibiotic, anti-inflammatory, gastroprotective, and antihistamine effect, making it suitable for treating complex conditions where infections are accompanied by inflammation and allergic reactions. The compound demonstrates good predicted binding affinities to target proteins responsible for bacterial cell wall synthesis, cyclooxygenase enzymes, and histamine receptors. Although promising, the development and deployment of such a sophisticated compound bear moderate toxicity risks involving vital organs such as the liver and kidneys. The compound is expected to have a rapid absorption profile, moderate distribution, followed by metabolism majorly in the liver and excretion through renal and fecal pathways. The future research direction leans towards optimizing the compound's safety profile, extending its indications, and verifying the therapeutic predictions through clinical trials.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>